UNIT - 3

Two Degree Freedom Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Formulation of equation of motion, natural frequencies and modes of vibration by classical method, coupled pendulum, forced vibration, dynamic vibration absorber.

1. Introduction to Two Degree Freedom Systems

A system needing two independent co-ordinates to describe its motion = 2-DOF system. Examples: two masses connected by springs, double pendulum, torsional system of two rotors, vehicle (bounce + pitch).

- Two equations of motion (coupled), two natural frequencies $\omega_1 < \omega_2$.
- For each natural frequency there is a definite amplitude ratio X_2/X_1 - called the principal (normal) mode of vibration.
- In a principal mode both masses move harmonically at the same frequency, in phase or 180° out of phase, passing through equilibrium together.
- Free vibration is the superposition of the two normal modes (depends on initial conditions).
- If excited at a natural frequency, the system vibrates in the corresponding mode only.
- Number of natural frequencies = number of DOF. Node = point of zero amplitude in a mode.

Types of coupling

- Static (elastic) coupling : stiffness matrix has off-diagonal term ($k_{12} \neq 0$) - displacement of one co-ordinate produces force on the other.
- Dynamic (mass / inertia) coupling : mass matrix has off-diagonal term ($m_{12} \neq 0$) - acceleration of one co-ordinate produces force on the other.
- Choice of co-ordinates changes coupling. Principal (normal) co-ordinates uncouple the equations completely (both M and K diagonal).

2. Formulation of Equations of Motion

Consider masses m_1, m_2, springs k_1 (wall - m_1), k_2 (between masses), k_3 (m_2 - wall).

Displacements x_1, x_2 from equilibrium.

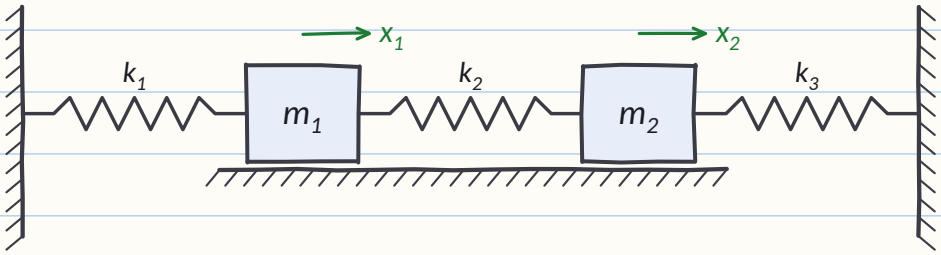


Fig 3.1 Two-DOF spring-mass system

Newton's second law for each mass:

$$m_1 \ddot{x}_1 = - k_1 x_1 + k_2(x_2 - x_1)$$
$$m_2 \ddot{x}_2 = - k_2(x_2 - x_1) - k_3 x_2 \quad (1)$$

Rearranged (coupled equations):

$$m_1 \ddot{x}_1 + (k_1 + k_2)\, x_1 - k_2\, x_2 = 0$$
$$m_2 \ddot{x}_2 - k_2\, x_1 + (k_2 + k_3)\, x_2 = 0 \quad (2)$$

Matrix form: $[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$

$$[\, m_1\ 0\ ;\ 0\ m_2\,]\ \{\ddot{x}_1, \ddot{x}_2\} + [\, k_1{+}k_2\ -k_2\ ;\ -k_2\ k_2{+}k_3\,]\ \{x_1, x_2\} = \{0, 0\} \quad (3)$$

- $[M]$ diagonal (no dynamic coupling), $[K]$ symmetric with off-diagonal $-k_2$ (static coupling). Always $K_{ij} = K_{ji}$ (Maxwell reciprocity).
- With damping: $[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F(t)\}$.

3. Natural Frequencies and Mode Shapes (Classical Method)

Assume harmonic solution, both masses at same frequency ω:

$$x_1 = X_1 \cos(\omega t + \varphi),\ x_2 = X_2 \cos(\omega t + \varphi) \quad (4)$$

Substitute in (2):

$$(k_1 + k_2 - m_1\omega^2)\, X_1 - k_2\, X_2 = 0$$
$$- k_2\, X_1 + (k_2 + k_3 - m_2\omega^2)\, X_2 = 0 \quad (5)$$

Non-trivial solution (X_1, X_2 ≠ 0) needs determinant of coefficients = 0 → frequency (characteristic) equation:

$$(k_1 + k_2 - m_1\omega^2)(k_2 + k_3 - m_2\omega^2) - k_2^2 = 0$$
$$m_1m_2\,\omega^4 - [\, m_1(k_2+k_3) + m_2(k_1+k_2)\,]\,\omega^2 + (k_1k_2 + k_2k_3 + k_3k_1) = 0 \quad (6)$$

Quadratic in ω^2: two positive real roots ω_1^2, ω_2^2 (natural frequencies; ω_1 is the fundamental).

Mode shapes (amplitude ratio) from eqn (5) for each root:

$$r_i = X_2/X_1\,|_{\omega\,=\,\omega i} = (k_1 + k_2 - m_1\omega_i^2) / k_2 = k_2 / (k_2 + k_3 - m_2\omega_i^2) \quad (7)$$

- Mode 1 (lower frequency): r_1 positive → masses move in phase, no node.
- Mode 2 (higher frequency): r_2 negative → masses move out of phase, one node between them.

General solution - superposition of the two modes (4 constants from initial conditions $x_1(0)$, $x_2(0)$, $\dot{x}_1(0)$, $\dot{x}_2(0)$):

$$x_1 = X_1^{(1)}\cos(\omega_1 t + \varphi_1) + X_1^{(2)}\cos(\omega_2 t + \varphi_2)$$
$$x_2 = r_1X_1^{(1)}\cos(\omega_1 t + \varphi_1) + r_2X_1^{(2)}\cos(\omega_2 t + \varphi_2) \quad (8)$$

Orthogonality of principal modes

Modal vectors $\{X\}^{(1)}$, $\{X\}^{(2)}$ satisfy: $\{X\}^{(1)T}\,[M]\,\{X\}^{(2)} = 0$ and $\{X\}^{(1)T}\,[K]\,\{X\}^{(2)} = 0$.

- Basis of modal analysis: with $x = [\Phi]\,q$, the equations decouple into two SDOF equations $M_i\,\ddot{q}_i + K_i\,q_i = 0$, $\omega_i^2 = K_i/M_i$.

Proof (sketch): for modes i, j: $[K]\{X\}_i = \omega_i^2[M]\{X\}_i$. Pre-multiply by $\{X\}_j^T$, and write the same with $i \leftrightarrow j$, subtract using symmetry of [M],[K]:

$$(\omega_i^2 - \omega_j^2)\,\{X\}_j^T[M]\{X\}_i = 0 \rightarrow \{X\}_j^T[M]\{X\}_i = 0 \text{ for } \omega_i \neq \omega_j \quad (8a)$$

Modal (normal-co-ordinate) analysis - steps

1. Find ω_1, ω_2 and mode vectors; form modal matrix $[\Phi] = [\,\{X\}^{(1)}\,\{X\}^{(2)}\,]$.
2. Transform $\{x\} = [\Phi]\{q\}$. Then $[\Phi]^T[M][\Phi] = \text{diag}(M_1, M_2)$, $[\Phi]^T[K][\Phi] = \text{diag}(K_1, K_2)$.
3. Solve two SDOF equations $M_i\ddot{q}_i + K_iq_i = Q_i(t)$; initial $q = [\Phi]^{-1}x(0)$.
4. Transform back $\{x\} = [\Phi]\{q\}$.

Matrix (eigenvalue) form

Eqn (5) is the eigenvalue problem $[K]\{X\} = \omega^2[M]\{X\}$ or $[D]\{X\} = (1/\omega^2)\{X\}$ with dynamic matrix $[D] = [K]^{-1}[M]$ (flexibility × mass) - used for iterative (Stodola) methods in Unit IV.

Special case : equal masses, equal springs ($k_1 = k_2 = k_3 = k$)

Frequency equation: $m^2\omega^4 - 4km\omega^2 + 3k^2 = 0 \rightarrow (m\omega^2 - k)(m\omega^2 - 3k) = 0$

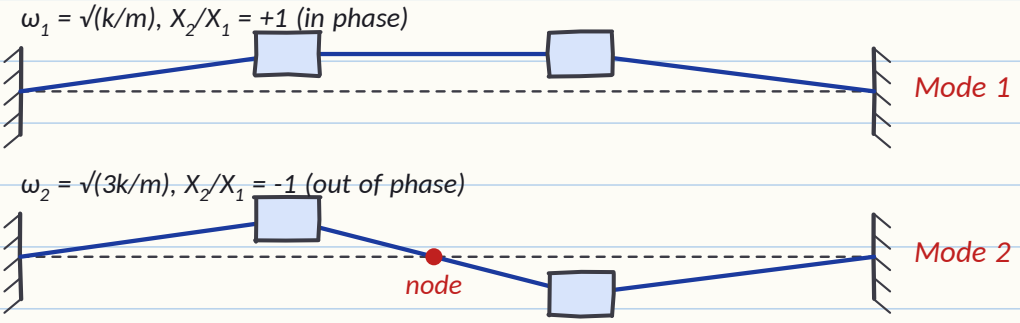


Fig 3.2 Mode shapes (equal masses and springs)

Mode 1: $X_2/X_1 = (2k - m\omega_1^2)/k = 1$; Mode 2: $X_2/X_1 = (2k - 3k)/k = -1$.

Problem 1 (golden-ratio system, free response)

$m_1 = m_2 = 10$ kg, $k_1 = k_2 = 4000$ N/m, $k_3 = 0$ (wall - k_1 - m_1 - k_2 - m_2). Find frequencies, modes; and response if m_1 is displaced 1 mm, m_2 held at 0, released.

Eqn (6) with $k_3 = 0$: $m^2\omega^4 - 3km\omega^2 + k^2 = 0 \rightarrow \omega^2 = (3 \pm \sqrt{5})/2 \cdot k/m = 0.382(400), 2.618(400)$

$\omega_1^2 = 152.8$, $\omega_1 = 12.36$ rad/s (1.97 Hz); $\omega_2^2 = 1047$, $\omega_2 = 32.36$ rad/s (5.15 Hz)

Modes: $r_1 = (2k - m\omega_1^2)/k = 2 - 0.382 = 1.618$; $r_2 = 2 - 2.618 = -0.618$

Initial $x = (1, 0)$: $c_1 + c_2 = 1$; $1.618c_1 - 0.618c_2 = 0 \rightarrow c_1 = 0.2764$, $c_2 = 0.7236$

$x_1 = 0.2764 \cos 12.36t + 0.7236 \cos 32.36t$ (mm)

$x_2 = 0.4472 \cos 12.36t - 0.4472 \cos 32.36t$ (mm)

Symmetric system with coupling spring ($k_1 = k_3 = k$, $k_2 = k_c$, $m_1 = m_2 = m$)

Add and subtract the equations → decoupled in $q_1 = x_1 + x_2$ (in phase) and $q_2 = x_1 - x_2$ (out of phase):

$$\omega_1 = \sqrt{(k/m)} \quad (X_2/X_1 = +1)$$
$$\omega_2 = \sqrt{((k + 2k_c)/m)} \quad (X_2/X_1 = -1) \qquad (8d)$$

Problem 1a (symmetric system)

$m = 5$ kg each; end springs $k = 3000$ N/m; coupling spring $k_c = 2000$ N/m. Find frequencies.

$\omega_1 = \sqrt{(3000/5)} = 24.5$ rad/s (3.90 Hz); $\omega_2 = \sqrt{((3000 + 4000)/5)} = \sqrt{1400} = 37.4$ rad/s (5.96 Hz)

Equations by Lagrange's method (preview of Unit IV)

$T = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2$, $U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2 + \tfrac{1}{2}k_3x_2^2$. Apply $d/dt(\partial T/\partial\dot{x}_i) - \partial T/\partial x_i + \partial U/\partial x_i = Q_i$:

$i = 1$: $m_1\ddot{x}_1 + k_1x_1 - k_2(x_2 - x_1) = 0$; $i = 2$: $m_2\ddot{x}_2 + k_2(x_2 - x_1) + k_3x_2 = 0$ - identical to eqn (2).

Problem 2 (unequal masses)

$m_1 = 2$ kg, $m_2 = 1$ kg, $k_1 = k_2 = 1000$ N/m, $k_3 = 0$. Find ω_1, ω_2 and modes.

Eqn (6): $2\omega^4 - [2(1000) + 1(2000)]\omega^2 + 10^6 = 0 \rightarrow 2\omega^4 - 4000\omega^2 + 10^6 = 0 \rightarrow \omega^2 =$ $1000(1 - 0.7071)$ and $1000(1 + 0.7071)$

$\omega_1 = 17.11$ rad/s (2.72 Hz), $\omega_2 = 41.32$ rad/s (6.58 Hz)

$r_1 = (2000 - 2\times292.9)/1000 = +1.414$; $r_2 = (2000 - 2\times1707.1)/1000 = -1.414$.

Problem 2a (torsional 2-DOF)

Rotor $J_1 = 2$ kg·m^2 fixed to wall by shaft $k_{t1} = 1000$ N·m/rad; rotor $J_2 = 1$ kg·m^2 connected to J_1 by shaft $k_{t2} = 1000$ N·m/rad. Find natural frequencies.

Mathematically identical to Problem 2 ($J \leftrightarrow m$, $k_t \leftrightarrow k$): $2\omega^4 - 4000\omega^2 + 10^6 = 0 \rightarrow$ $\omega_1 = 17.1$, $\omega_2 = 41.3$ rad/s; $\theta_2/\theta_1 = +1.414$ and -1.414.

Double pendulum (classical 2-DOF example)

Two equal masses m on massless rods of length l hanging one below the other; small angles θ_1, θ_2 (Lagrange or Newton):

$$2\,l\,\ddot{\theta}_1 + l\,\ddot{\theta}_2 + 2\,g\,\theta_1 = 0$$
$$l\,\ddot{\theta}_1 + l\,\ddot{\theta}_2 + g\,\theta_2 = 0 \qquad (8b)$$

With $\lambda = \omega^2 l/g$: $(2 - 2\lambda)\Theta_1 - \lambda\Theta_2 = 0$, $-\lambda\Theta_1 + (1 - \lambda)\Theta_2 = 0 \rightarrow \lambda^2 - 4\lambda + 2 = 0$

$$\omega^2 = (2 - \sqrt{2})\,g/l \text{ and } (2 + \sqrt{2})\,g/l \rightarrow \omega_1 = 0.765\,\sqrt{(g/l)},\ \omega_2 = 1.848\,\sqrt{(g/l)}$$
$$\Theta_2/\Theta_1 = (2 - 2\lambda)/\lambda = +1.414 \text{ (mode 1)}, -1.414 \text{ (mode 2)} \qquad (8c)$$

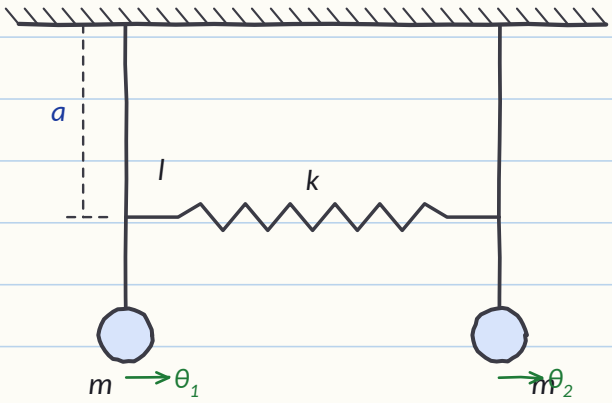
For $l = 1$ m: $\omega_1 = 2.40$ rad/s, $\omega_2 = 5.79$ rad/s. In mode 1 both swing the same way (lower bob moves more); in mode 2 they swing opposite, with a node on the upper rod.

Torsional two-rotor, three-segment systems

Same method with $m \rightarrow J$ (inertia), $k \rightarrow k_t = GJ/l$, $x \rightarrow \theta$: $[J]\{\ddot{\theta}\} + [k_t]\{\theta\} = 0$. If a shaft end is free (no restraint) one root is $\omega = 0$ (rigid-body rotation); semi-definite system. e.g. two rotors: $\omega_n = \sqrt{(k_t(J_1 + J_2)/(J_1J_2))}$ (Unit I).

4. Coupled Pendulum

Two identical simple pendulums (mass m, length l) joined by a spring k attached at distance a below pivot. Small angles θ_1, θ_2.



$$m l^2 \ddot{\theta}_1 + m g l \theta_1 + k a^2(\theta_1 - \theta_2) = 0$$
$$m l^2 \ddot{\theta}_2 + m g l \theta_2 - k a^2(\theta_1 - \theta_2) = 0$$

Fig 3.3 Coupled pendulum

Equations of motion (moments about pivot, small θ):

$$m l^2 \ddot{\theta}_1 + (m g l + k a^2)\theta_1 - k a^2 \theta_2 = 0$$
$$m l^2 \ddot{\theta}_2 - k a^2 \theta_1 + (m g l + k a^2)\theta_2 = 0 \quad (9)$$

Add and subtract (principal co-ordinates $q_1 = \theta_1 + \theta_2$, $q_2 = \theta_1 - \theta_2$):

$$\ddot{q}_1 + (g/l) q_1 = 0 \rightarrow \omega_1 = \sqrt{(g/l)} \text{ (in phase; spring not stretched)}$$
$$\ddot{q}_2 + (g/l + 2ka^2/ml^2) q_2 = 0 \rightarrow \omega_2 = \sqrt{(g/l + 2ka^2/ml^2)} \text{ (out of phase)} \quad (10)$$

- Mode 1: $\theta_1 = \theta_2$ - pendulums swing together at the free-pendulum frequency.
- Mode 2: $\theta_1 = -\theta_2$ - spring stretches/compresses; higher frequency. If a = l (spring at bob): $\omega_2^2 = g/l + 2k/m$.

Beating - energy transfer

Start with $\theta_1(0) = \theta_0$, $\theta_2(0) = 0$, released from rest: $\theta_1 = (\theta_0/2)(\cos \omega_1 t + \cos \omega_2 t)$, $\theta_2 = (\theta_0/2)(\cos \omega_1 t - \cos \omega_2 t)$

$$\theta_1 = \theta_0 \cos[(\omega_2 - \omega_1)t/2] \cos[(\omega_2 + \omega_1)t/2]$$
$$\theta_2 = \theta_0 \sin[(\omega_2 - \omega_1)t/2] \sin[(\omega_2 + \omega_1)t/2] \quad (11)$$

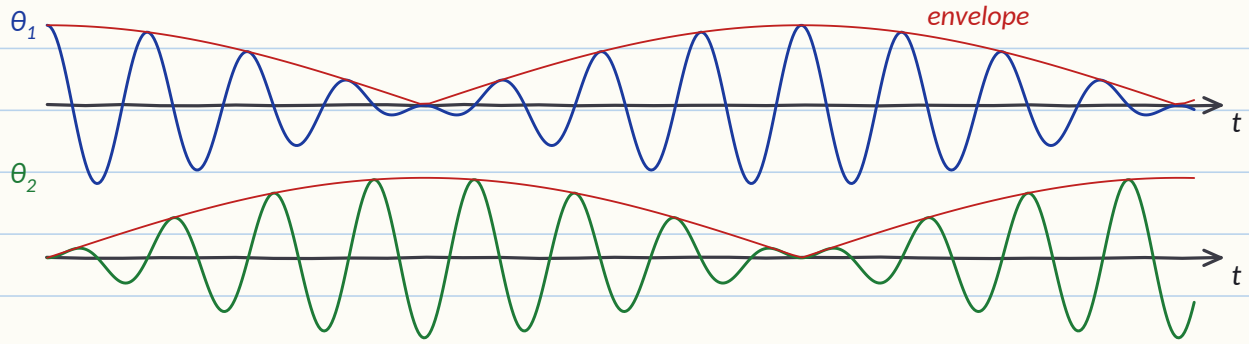


Fig 3.4 Beats: energy passes back and forth between pendulums

- *Pendulum 1 loses amplitude as 2 gains; complete transfer takes $t = \pi/(\omega_2 - \omega_1)$. Weak coupling (small k) → slow beats.*

Problem 3 (coupled pendulum)

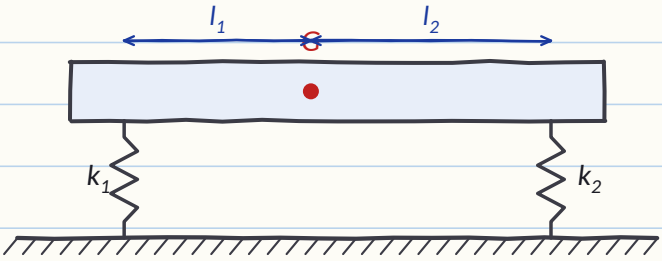
l = 1 m, m = 1 kg, spring k = 2 N/m attached at the bobs ($a = l$). Find ω_1, ω_2 and time for complete energy transfer.

$\omega_1 = \sqrt{(9.81/1)} = 3.132$ rad/s; $\omega_2 = \sqrt{(9.81 + 2\times2/1)} = \sqrt{13.81} = 3.716$ rad/s

$t = \pi/(\omega_2 - \omega_1) = \pi/0.584 = 5.38$ s.

5. Coordinate Coupling - Vehicle Bounce and Pitch

Rigid car body (mass m, inertia J about C.G.) on front and rear suspensions k_1, k_2 at distances l_1, l_2 from C.G. Co-ordinates: x (bounce of C.G.), θ (pitch).



$$m\ddot{x} + (k_1+k_2)x - (k_1l_1 - k_2l_2)\theta = 0$$

$$J\ddot{\theta} - (k_1l_1 - k_2l_2)x + (k_1l_1^2 + k_2l_2^2)\theta = 0$$

coupling term: $k_1l_1 - k_2l_2$

Fig 3.5 Bounce-pitch model

- *If $k_1l_1 = k_2l_2$ the equations are uncoupled (static coupling vanishes): $\omega_x = \sqrt{((k_1+k_2)/m)}$, $\omega_\theta = \sqrt{((k_1l_1^2+k_2l_2^2)/J)}$.*
- *Otherwise bounce and pitch always occur together; the system oscillates about two nodes (centres of oscillation). Poor ride comfort when node lies inside the wheelbase.*
- *Coupling depends on chosen co-ordinates; using co-ordinates measured at a point other than G can give dynamic coupling ($m_{12} \neq 0$) instead.*

Problem 4 (uncoupled case)

$m = 1200$ kg, $J = 1500$ kg·m^2, $l_1 = 1.2$ m, $l_2 = 1.6$ m, $k_1 = 24$ kN/m, $k_2 = 18$ kN/m. Check coupling, find frequencies.

$k_1 l_1 = 28\,800 = k_2 l_2 = 28\,800 \rightarrow$ uncoupled. $\omega_x = \sqrt{(42000/1200)} = 5.92$ rad/s (0.94 Hz)

$\omega_\theta = \sqrt{((24000\times1.44 + 18000\times2.56)/1500)} = \sqrt{53.76} = 7.33$ rad/s (1.17 Hz)

6. Forced Vibration of Two-DOF System (undamped)

Harmonic force $F_1 = F_0 \sin \omega t$ on mass m_1; none on m_2.

$$m_1 \ddot{x}_1 + (k_1 + k_2)\, x_1 - k_2\, x_2 = F_0 \sin \omega t$$
$$m_2 \ddot{x}_2 - k_2\, x_1 + (k_2 + k_3)\, x_2 = 0 \qquad (12)$$

Steady state: $x_i = X_i \sin \omega t$:

$$(k_1+k_2 - m_1\omega^2)\, X_1 - k_2\, X_2 = F_0$$
$$-k_2\, X_1 + (k_2+k_3 - m_2\omega^2)\, X_2 = 0 \qquad (13)$$

Cramer's rule with $\Delta(\omega) = (k_1+k_2 - m_1\omega^2)(k_2+k_3 - m_2\omega^2) - k_2^2$:

$$X_1 = (k_2 + k_3 - m_2\omega^2)\, F_0 / \Delta(\omega)$$
$$X_2 = k_2\, F_0 / \Delta(\omega) \qquad (14)$$

- $\Delta(\omega) = 0$ is the frequency equation $\rightarrow$ two resonances, at $\omega = \omega_1$ and $\omega = \omega_2$.
- If $k_2 + k_3 - m_2\omega^2 = 0 \rightarrow X_1 = 0$ (anti-resonance): m_1 stands still although it carries the force! Basis of the vibration absorber.
- Between the two resonances the masses move out of phase; beyond ω_2 amplitudes fall. Phase changes by 180° across each resonance.

Problem 6 (forced response, numbers)

Use the system of Problem 1 ($m_1 = m_2 = 10$ kg, $k_1 = k_2 = 4000$ N/m, $k_3 = 0$) with $F_0 = 20$ N on m_1. Find X_1, X_2 at $\omega = 10$ and $\omega = 20$ rad/s.

$\Delta(\omega) = (8000 - 10\omega^2)(4000 - 10\omega^2) - 4000^2$

$\omega = 10$: $\Delta = (7000)(3000) - 1.6\times10^7 = 5\times10^6$; $X_1 = (3000)(20)/\Delta = 12$ mm, $X_2 = 4000(20)/\Delta = 16$ mm (in phase, below $\omega_1 = 12.36$).

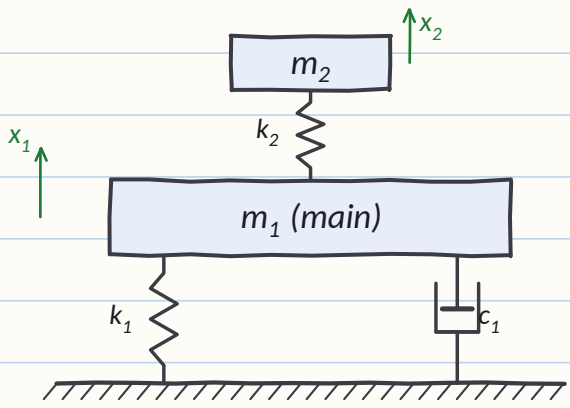
$\omega = 20$: $4000 - 10(400) = 0 \rightarrow X_1 = 0$ (anti-resonance), $X_2 = k_2 F_0/\Delta = 4000(20)/(-1.6\times10^7) = -5$ mm: m_2 acts as an absorber; its spring force $k_2 X_2 = -20$ N $= -F_0$.

Damped forced response (outline)

- Equations: $[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F\}e^{i\omega t} \rightarrow \{X\} = [K - \omega^2 M + i\omega C]^{-1}\{F\}$ (matrix of mechanical impedance $Z(\omega) = K - \omega^2 M + i\omega C$).
- Damping removes the infinite peaks at ω_1, ω_2 and fills the anti-resonance ($X_1 \neq 0$ exactly). With proportional damping $[C] = \alpha[M] + \beta[K]$ the modal method still decouples.

7. Dynamic Vibration Absorber (DVA)

A machine (m_1, k_1) is excited by force $F_0 \sin \omega t$ at (or near) its resonance → large vibration. Attach an auxiliary spring-mass (m_2, k_2) - the dynamic vibration absorber - to cancel it. Becomes a 2-DOF system.



Absorber tuned: $\omega_a = \sqrt{(k_2/m_2)} = \omega$

Main mass: $X_1 = 0$

Absorber: $X_2 = -F_0/k_2$

Fig 3.6 Dynamic vibration absorber

Equations (main system with $k_3 = 0$, damping neglected) - same as (12) with $k_1 \rightarrow k_1$, k_2, and no k_3:

$$m_1 \ddot{x}_1 + (k_1 + k_2) x_1 - k_2 x_2 = F_0 \sin \omega t$$
$$m_2 \ddot{x}_2 + k_2(x_2 - x_1) = 0 \qquad (15)$$

From (14) with $k_3 = 0$:

$$X_1 = (k_2 - m_2\omega^2) F_0 / [\, (k_1+k_2 - m_1\omega^2)(k_2 - m_2\omega^2) - k_2^2 \,]$$
$$X_2 = k_2 F_0 / [\text{ same denominator }] \qquad (16)$$

Tuned absorber : $k_2 - m_2\omega^2 = 0$, i.e. $\omega_a = \sqrt{(k_2/m_2)} = \omega$ (absorber natural frequency = excitation frequency):

$$X_1 = 0,\ X_2 = -F_0/k_2$$

spring force on main mass: $k_2X_2 = -F_0$ - exactly cancels exciting force. (17)

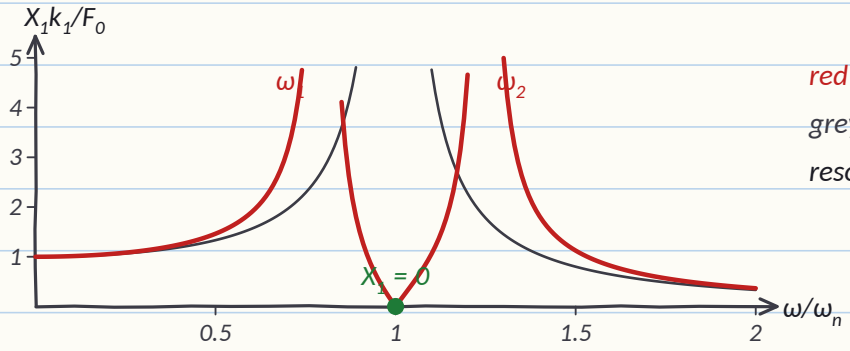
- Design: choose $k_2 = F_0/X_{2,allowed}$ (limits absorber stroke); then $m_2 = k_2/\omega^2$.
- Mass ratio $\mu = m_2/m_1$. Typical μ = 0.05-0.25. Absorber should not be so light that its stroke is excessive.

Natural frequencies after adding absorber ($\omega_a = \omega_n = \sqrt{(k_1/m_1)}$)

Let $s = (\omega/\omega_n)^2$, $k_2/k_1 = \mu$. Denominator of (16) = 0 → $s^2 - (2 + \mu)s + 1 = 0$

$$(\omega/\omega_n)^2 = 1 + \mu/2 - \sqrt{(\mu + \mu^2/4)} \text{ and } 1 + \mu/2 + \sqrt{(\mu + \mu^2/4)} \quad (18)$$

- Original resonance at ω_n is replaced by two resonances, one below and one above ω_n. Larger μ → wider separation → wider operating band.
- Drawback: at speeds other than the tuned speed (e.g. while starting/stopping) the machine passes through one of the new resonances; works for narrow speed range.



red : with absorber (μ = 0.2)

grey: without absorber

resonances at 0.80 ω_n, 1.25 ω_n

Fig 3.7 Main-mass response with tuned absorber

Problem 5 (design of absorber)

Machine m_1 = 200 kg, k_1 = 80 kN/m, force 100 N at ω = 20 rad/s (resonance, $\omega_n = \sqrt{(80000/200)} = 20$). Absorber stroke limited to 5 mm. Find k_2, m_2, new natural frequencies.

$k_2 = F_0/X_2 = 100/0.005 =$ 20 000 N/m; $m_2 = k_2/\omega^2 = 20000/400 =$ 50 kg (μ = 0.25, ω_a = 20 rad/s)

$(\omega/\omega_n)^2 = 1.125 - 0.515 = 0.610$ and $1.125 + 0.515 = 1.640$

$\omega_1 = 20\sqrt{0.610} = 15.6$ rad/s; $\omega_2 = 20\sqrt{1.640} = 25.6$ rad/s - operate outside 15.6-25.6 range except at tuned speed.

Damped absorber (Den Hartog optimum)

- Absorber with viscous damper c_2: amplitude of main mass remains finite over a wide frequency band (no zero, but no infinite peaks).
- Optimum tuning: $f_{opt} = \omega_a/\omega_n = 1/(1 + \mu)$; optimum damping: $\zeta_{opt} = \sqrt{[\ 3\mu / (8(1 + \mu)^3)\ ]}$.
- Examples: pendulum absorbers in skyscrapers (tuned mass dampers), engine crankshaft torsional dampers, power-line stockbridge dampers, bladed disks.

Applications of 2-DOF theory

- Vehicle suspension (quarter-car: sprung + unsprung mass), bounce-pitch ride analysis.
- Tuned mass dampers on tall buildings and bridges; crankshaft torsional dampers; stockbridge dampers on transmission lines.
- Machine tool: spindle + tool holder; rotor-bearing systems; two-storey shear building (m_1, m_2, storey stiffnesses k_1, k_2).
- Vibration absorbers for helicopters, washing machines, ship propeller shafts.

8. Quick Revision & Short Questions

Item	Result
Equation of motion	$[M]\{\ddot{x}\} + [K]\{x\} = 0$; assume $x = X\cos(\omega t + \varphi)$
Frequency equation	$\det[K - \omega^2 M] = 0 \rightarrow$ quadratic in ω^2
Mode shape ratio	$X_2/X_1 = (k_1+k_2 - m_1\omega^2)/k_2$
Equal m, k (3 springs)	$\omega_1 = \sqrt{(k/m)}$ (1 : 1); $\omega_2 = \sqrt{(3k/m)}$ (1 : -1)
Coupled pendulum	$\omega_1 = \sqrt{(g/l)}$; $\omega_2 = \sqrt{(g/l + 2ka^2/ml^2)}$
Anti-resonance	$X_1 = 0$ when $\omega^2 = k_2/m_2$
DVA	$\omega_a = \omega$; $X_2 = -F_0/k_2$; new ω from $s^2 - (2+\mu)s + 1 = 0$

1. What is a principal mode? A vibration in which every point moves harmonically at the same frequency, with fixed amplitude ratio.
2. What is a node? A point of zero displacement in a mode; mode 2 of a 2-DOF system has one node.
3. Static vs dynamic coupling? Off-diagonal terms in [K] vs [M]; depends on co-ordinates chosen.

4. **What are normal co-ordinates?** Co-ordinates in which both [M] and [K] become diagonal; each equation is a SDOF equation.
5. **Why does a DVA work only near one frequency?** It is tuned to $\omega_a = \omega$; at other frequencies absorber force no longer cancels F_0.
6. **How many natural frequencies does a 2-DOF have?** Two; semi-definite systems have one zero (rigid-body) frequency.
7. **Why are modes orthogonal?** Because [M],[K] are symmetric and $\omega_i \neq \omega_j$; $\{X\}_i^T[M]\{X\}_j = 0$ - each mode can be treated independently.
8. **What is a semi-definite system?** One with a rigid-body mode ($\omega = 0$), e.g. two rotors on a free shaft, a train of cars.
9. **Effect of mass ratio on DVA?** Larger μ widens the band between the two new resonances (more robust) but adds more mass.
10. **What happens at anti-resonance?** The main mass has zero steady amplitude; absorber spring force balances the exciting force.

Important questions: (1) derive natural frequencies & modes of 2-DOF spring-mass system (2) coupled pendulum with beats (3) forced response & anti-resonance (4) design a dynamic vibration absorber (5) bounce-pitch coupling of a vehicle.